

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times int \times int \times int \Rightarrow State \\ \text{noncrit0} : Loc \\ \text{level0} : Loc \\ \text{crit0} : Loc \\ \text{noncrit1} : Loc \\ \text{level1} : Loc \\ \text{crit1} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit0}, \text{proc1}, p0, p1, y) \rightarrow \text{state}(\text{level0}, \text{proc1}, p01, p1, y1) \quad [p01 = 1 \wedge y1 = 0] \\ 2 : & \text{state}(\text{level0}, \text{proc1}, p0, p1, y) \rightarrow \text{state}(\text{crit0}, \text{proc1}, p0, p1, y) \quad [y \neq 0 \vee (p1 < 1)] \\ 3 : & \text{state}(\text{crit0}, \text{proc1}, p0, p1, y) \rightarrow \text{state}(\text{noncrit0}, \text{proc1}, p01, p1, y) \quad [p01 = 0] \\ 4 : & \text{state}(\text{proc0}, \text{noncrit1}, p0, p1, y) \rightarrow \text{state}(\text{proc0}, \text{level1}, p0, p11, y1) \quad [p11 = 1 \wedge y1 = 1] \\ 5 : & \text{state}(\text{proc0}, \text{level1}, p0, p1, y) \rightarrow \text{state}(\text{proc0}, \text{crit1}, p0, p1, y) \quad [y \neq 1 \vee (p0 < 1)] \\ 6 : & \text{state}(\text{proc0}, \text{crit1}, p0, p1, y) \rightarrow \text{state}(\text{proc0}, \text{noncrit1}, p0, p11, y) \quad [p11 = 0] \\ 7 : & \text{state}(\text{proc0}, \text{proc1}, p0, p1, y) \rightarrow \text{success} \\ 8 : & \text{state}(\text{crit0}, \text{crit1}, p0, p1, y) \rightarrow \text{error} \end{array} \right.$$

$\{g: \text{state}(\text{noncrit0}, \text{noncrit1}, 0, 0, g) \Rightarrow \text{success}\}$

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bitvec_1 \times bitvec_1 \times bitvec_1 \Rightarrow State \\ \text{noncrit0} : Loc \\ \text{level0} : Loc \\ \text{crit0} : Loc \\ \text{noncrit1} : Loc \\ \text{level1} : Loc \\ \text{crit1} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll}
1 : & \text{state}(\text{noncrit0}, \text{proc1}, p0, p1, y) \rightarrow \text{state}(\text{level0}, \text{proc1}, p01, p1, y1) \quad [p01 = \#b1 \wedge y1 = \#b0] \\
2 : & \text{state}(\text{level0}, \text{proc1}, p0, p1, y) \rightarrow \text{state}(\text{crit0}, \text{proc1}, p0, p1, y) \quad [y \neq \#b0 \vee (p1 \text{bvult} \#b1)] \\
3 : & \text{state}(\text{crit0}, \text{proc1}, p0, p1, y) \rightarrow \text{state}(\text{noncrit0}, \text{proc1}, p01, p1, y) \quad [p01 = \#b0] \\
4 : & \text{state}(\text{proc0}, \text{noncrit1}, p0, p1, y) \rightarrow \text{state}(\text{proc0}, \text{level1}, p0, p11, y1) \quad [p11 = \#b1 \wedge y1 = \#b1] \\
5 : & \text{state}(\text{proc0}, \text{level1}, p0, p1, y) \rightarrow \text{state}(\text{proc0}, \text{crit1}, p0, p1, y) \quad [y \neq \#b1 \vee (p0 \text{bvugt} \#b1)] \\
6 : & \text{state}(\text{proc0}, \text{crit1}, p0, p1, y) \rightarrow \text{state}(\text{proc0}, \text{noncrit1}, p0, p11, y) \quad [p11 = \#b0] \\
7 : & \text{state}(\text{proc0}, \text{proc1}, p0, p1, y) \rightarrow \text{success} \\
8 : & \text{state}(\text{crit0}, \text{crit1}, p0, p1, y) \rightarrow \text{error}
\end{array} \right.$$

$\{9: \text{state}(\text{noncrit0}, \text{noncrit1}, \#b0, \#b0, y) \Rightarrow \text{success}\}$